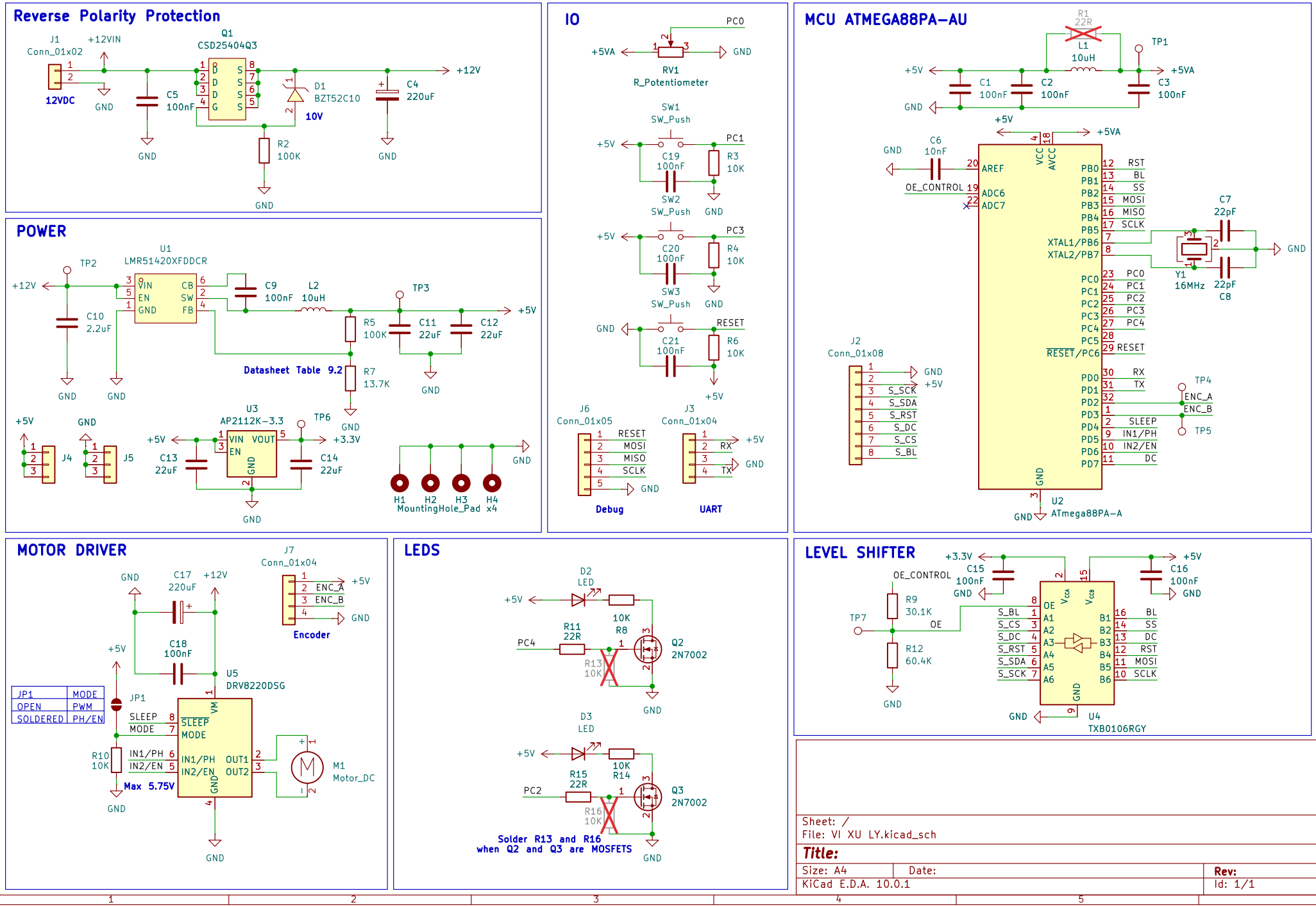

Reverse Polarity Protection
J1
Conn_01x02
12VDC
+12VIN
GND
C5
100nF
Q1
CSD25404Q3
D1
BZT52C10
10V
C4
220uF
+12V
R2
100K
POWER
U1
LMR51420XFDDCR
VIN
EN
GND
CB
SW
FB
TP2
C10
2.2uF
C9
100nF
L2
10uH
TP3
R5
100K
C11
22uF
C12
22uF
+5V
Datasheet Table 9.2
R7
13.7K
TP6
U3
AP2112K-3.3
VIN
VOUT
EN
GND
+3.3V
C13
22uF
C14
22uF
J4
J5
H1
H2
H3
H4
MountingHole_Pad x4
IO
PC0
RV1
R_Potentiometer
SW1
SW_Push
PC1
C19
100nF
R3
10K
SW2
SW_Push
PC3
C20
100nF
R4
10K
SW3
SW_Push
RESET
C21
100nF
R6
10K
J6
Conn_01x05
RESET
MOSI
MISO
SCLK
Debug
J3
Conn_01x04
RX
TX
UART
MCU ATMEGA88PA-AU
R1
22R
L1
10uH
TP1
C1
100nF
C2
100nF
C3
100nF
C6
10nF
OE_CONTROL
AREF
ADC6
ADC7
VCC
AVCC
PB0
PB1
PB2
PB3
PB4
PB5
XTAL1/PB6
XTAL2/PB7
PC0
PC1
PC2
PC3
PC4
PC5
RESET/PC6
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
RST
BL
SS
MOSI
MISO
SCLK
C7
22pF
Y1
16MHz
C8
J2
Conn_01x08
S_SCK
S_SDA
S_RST
S_DC
S_CS
S_BL
TP4
ENC_A
ENC_B
TP5
SLEEP
IN1/PH
IN2/EN
DC
U2
ATmega88PA-A
MOTOR DRIVER
J7
Conn_01x04
ENC_A
ENC_B
Encoder
C17
220uF
C18
100nF
U5
DRV8220DSG
JP1
JP1 MODE
OPEN PWM
SOLDERED PH/EN
SLEEP
MODE
IN1/PH
IN2/EN
VM
OUT1
OUT2
R10
10K
Max 5.75V
M1
Motor_DC
LEDS
D2
LED
R11
22R
10K
R8
Q2
2N7002
R13
10K
D3
LED
R15
22R
10K
R14
Q3
2N7002
R16
10K
PC4
PC2
Solder R13 and R16
when Q2 and Q3 are MOSFETS
LEVEL SHIFTER
+3.3V
C15
100nF
C16
100nF
OE_CONTROL
R9
30.1K
R12
60.4K
TP7
OE
S_BL
S_CS
S_DC
S_RST
S_SDA
S_SCK
A1
A2
A3
A4
A5
A6
B1
B2
B3
B4
B5
B6
U4
TXB0106RGY
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Date:
Rev:
KiCad E.D.A. 10.0.1
Id: 1/1